

Promit Sarker

Programmer, Researcher

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[Portfolio](#) | [LinkedIn](#) | [Github](#)

CAREER SUMMARY

AI Engineer focused on LLM systems, RAG pipelines, and applied machine learning.
Built and deployed AI solutions using LangChain, vector databases, and fine-tuned LLMs for real-world applications including healthcare, OCR-based QA, and task automation.

WORK EXPERIENCES

SM Technology | AI Engineer | May 2025 - Present

- Built and deployed production AI systems using RAG, LLM agents, and vector search (FAISS/ChromaDB) to automate client business workflows via LangChain and API tool calling.
- Evaluated and optimized proprietary and open-source LLMs for latency, accuracy, and cost, delivering scalable pipelines that replaced manual processes in real deployments.

Raim Laboratory | Jr. Software Developer | July 2023 - July 2024

- Designed and implemented a production-grade inventory and medicine management platform for a medical laboratory, featuring real-time stock tracking and supply chain monitoring to replace manual record-keeping workflows.
- Developed custom reporting modules and integrated lightweight AI-driven automation for data processing and anomaly detection, improving decision support and resource allocation.

SKILLS

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| <ul style="list-style-type: none">• Python<ul style="list-style-type: none">◦ Django, DRF, FastAPI• SQL<ul style="list-style-type: none">◦ PostgreSQL• Postman• Git | <ul style="list-style-type: none">• NoSQL<ul style="list-style-type: none">◦ MongoDB• Docker• Vector Database<ul style="list-style-type: none">◦ FAISS◦ Chromadb | <ul style="list-style-type: none">• AI<ul style="list-style-type: none">◦ MCP, RAG, Langchain, n8n◦ Vector Embedding• Machine Learning• Deep Learning• n8n• Fine Tuning LLM |
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RELEVANT EXPERIENCES

NSU HCI Design Inclusion and Access Lab | Research Assistant | 2023 - 2024

- As a Research Assistant at HCI DIAL, my team and I contributed to projects focused on refining user interfaces and enhancing user experience.
- I was actively involved in user studies and assisted in developing a wearable device to monitor health status. We aimed to harness technological advancements to support individuals living below the poverty line.

IEEE NSU Student Branch | Member | 2020 - 2021

- As a committed member of the IEEE NSU Student Branch, I collaborated with my teammates to organize and promote engaging seminars. I helped my fellow volunteers enhance outreach initiatives, ensuring that participants had a successful and informative experience.

RELEVANT PROJECTS

MedQ-Llama: A fine tuned LLM

[Github](#)

- Fine-tuned TinyLlama-1.1B-Chat using QLoRA on a custom medical dataset to build a domain-specific medical Q&A assistant.
- Built a full pipeline including data cleaning, QA pair generation, and 4-bit parameter-efficient fine-tuning. Produced a specialized LLM capable of medical knowledge retrieval and question answering.

BanglaBuddy: Bangla book based AI

[Github](#)

- Built an OCR-to-LLM pipeline for Bangla-language book QA using Retrieval-Augmented Generation.
- Implemented text cleaning, vector embedding, and semantic search with a vector database.
- Enabled natural language querying over scanned books using an LLM-based response system.

EAI-Commerce: an AI based E-commerce

[Github](#)

- Built an AI-powered conversational shopping assistant with product recommendation and multilingual support.
- Integrated ChromaDB-based semantic search and conversation memory for intelligent product discovery. And connected external APIs for real-time inventory and product retrieval.

Quantum for Mars

[Github](#)

- Implemented a Quantum ResNet (QResNet) model for Martian terrain classification in an emulated environment. Which enables it to run on Quantum Computers.
- Designed Quantum encoding pipelines using Basis and Amplitude Encoding.
- Achieved >80% classification accuracy on Curiosity's terrain datasets.

Luna: The daily AI Assistant

[Github](#)

- Engineered an AI task automation system using Whisper and GPT models for voice-to-task processing.
- Built a contextual chatbot with task-aware memory and multilingual command understanding.
- Enabled structured task extraction including priorities and due dates from natural language speech.

AbegiLLama : A fine tuned Bengali LLM

[Github](#)

- Fine-tuned Meta Llama-3-8B using Hugging Face Transformers and PEFT (LoRA) for Bengali conversational tasks, building end-to-end pipelines for dataset curation, tokenization, and supervised training to improve response quality over the base model.
- Optimized the model for Kaggle free-tier GPU execution and deployed it publicly on [Hugging Face Hub](#), enabling accessible experimentation and reproducible research.

OTHER EXPERIENCES

Spellbound - স্পেলবাইন্ড | Editorial Member | 2020 - 2022

- Spellbound, a Harry Potter magazine, was the online-based sole publication for Bengali Harry Potter enthusiasts.
- As an editorial team member, I contributed by ensuring the quality of the content, conducting peer reviews, and promoting the magazine alongside my fellow team members.

Manabu Bookstore | Co-Owner | 2020 - 2022

- Founded and ran a profitable online book-selling venture offering affordable, quality books; currently on break with plans to relaunch as a physical book café

Notre Dame Science Club | President | 2018 - 2019

- Organized inter-college science and cultural festivals promoting innovation and collaboration.

EDUCATION

North South University

2020 - 2025

Bachelor of Computer Science and Engineering